

Topic 18C: Organic Synthesis
18. be able to deduce the empirical formulae, molecular formulae and structural formulae of compounds from data obtained from combustion analysis, elemental percentage composition, characteristic reactions of functional groups, infrared spectra, mass spectra and nuclear magnetic resonance
19. be able to plan reaction schemes, of up to four steps, to form both familiar and unfamiliar compounds
20. understand methods of increasing the length of the carbon chain in a molecule by the use of magnesium to form Grignard reagents and the reactions of the latter with carbon dioxide and with carbonyl compounds in dry ether
21. be able to select and justify suitable practical procedures for carrying out reactions involving compounds with functional groups included in the specification, including identifying appropriate control measures to reduce risk, based on data about hazards
22. understand the following techniques used in the preparation and purification of organic compounds: i refluxing ii purification by washing iii solvent extraction iv recrystallisation v drying vi distillation, including steam distillation vii melting temperature determination viii boiling temperature determination
CORE PRACTICAL 16: The preparation of aspirin

Topic 19: Modern Analytical Techniques II

In order to develop their practical skills, students should be encouraged to carry out a range of practical experiments related to this topic. Possible experiments include using chromatography to identify unknowns such as metals in a 'silver' coin or amino acids in a protein.

Mathematical skills that could be developed in this topic include calculating R_f values, interpreting a variety of different spectra, calculating relative molecular masses, using the $(n+1)$ rule for proton NMR.

Within this topic, students can consider a wider range of instrumental methods used for analysis, such as NMR; and see how this technique is used in medicine through MRI scans. They can also see a wide range of applications that rely on a combination of different analytical techniques.